Prince Sultan University
College of Humanities and Sciences Dept. of Math and Sciences
2nd Term 2023- 2024

INSTITUTIONAL COURSE SYLLABUS TEMPLATE

Course Code: Math 111	Course Title: Calculus I
Course Instructor:	Email:
Credit Hours: 3 credit hours	Course Location: Various on Campus
Scheduled Office Hours:	
Office Location: Building 1,	Office
Co-Requisite: None	Pre-Requisite: Math 002
On Campus	

Announcement Procedure: for announcements either regarding exams or extra classes or class cancellation or any other important activity, you will receive notifications, via **MOODLE Announcement.**

Mission Statement

PSU Mission:

Prince Sultan University aims to provide the Middle East with quality education to the highest international standards. In its efforts towards successful and responsible life-long learning, PSU integrates modern technology, pedagogy, and human values for the advancement of scientific research, productivity, and leadership towards a more meaningful role in society. PSU is committed to effective management of institutional resources to optimize its multiple roles as a catalyst for new learning opportunities, national and international partnerships, continuous studies, professional growth, community service, and diversity in educational horizons for the good of humanity.

CHS Mission:

The College of Humanities and Sciences aspires to provide students with a high-quality education in the fields of humanities and sciences in order to advance their scientific knowledge, enhance their cultural awareness, broaden their horizons, and create a balance in student life that supports educational growth and life-long learning skills. CHS is also committed to producing quality research with distinct societal impact.

I. Course Description:

This course is designed to develop the topics of differential calculus. Emphasis is placed on limits, continuity and derivatives of algebraic and transcendental functions of one variable. The goal of this course is developing the student's geometric insight into the concepts of differentiation, and applying these concepts to problem solving and "real world application

II. Course Learning Outcomes: On the successful completion of this course, students will be able to demonstrate the following:

Skills	Course Learning Outcomes
Knowledge & Understanding	None
Skills	CLO1: Calculate the limit of a function using appropriate techniques including L'Hopital's rule (ILO 3, ILO 7) CLO2: Compute the derivative of functions by applying the rules of differentiation. (ILO 3, ILO 4) CLO3: Determine whether a function is continuous and/or differentiable (ILO 7) CLO4: Find extreme values of functions. (ILO 6) CLO6: Solve various real-world problems using calculus concepts. (ILO 7)
Values	CLO5: Apply the concept of differentiation to sketch the graph of a function in a systematic way. (ILO 3)

III. Tentative Weekly Course Schedule:

Week No.	Dates	UNIT/ TOPIC	CLO(s) alignment	STUDENT TASKS (assignments, exams, quizzes, presentations)	No. of Contact hours
1	Jun 23- 27	1.1 Four Ways to Represent a Function 1.2 Mathematical Models: A Catalog of Essential Functions 1.3 New Functions from Old Functions	NA	**Revisions** **Quiz 1** Sunday-30 (From 12:00 Pm to 12:30 Pm)	4
2	Jun 30- Jul 4	1.4 Exponential Functions 1.5 Inverse Functions and Logarithms 2.2 The Limit of a Function 2.3 Calculating Limits Using the Limit Laws	NA	**Quiz 2** Thursday-4 (From 12:00 Pm to 12:30 Pm)	4
3	Jul	2.5 Continuity	CLO1		3

	7-11	2.6 Limits at Infinity; Horizontal Asymptotes 2.7 Derivatives and Rate of Change 2.8 The Derivative as a Function		**Major I** Thursday-11 (From 12:00 Pm to 1:00 Pm)	
4	Jul 14-18	3.1 Derivative of Polynomials and Exponential Functions 3.2 The Product and Quotient Rules 3.3 Derivatives of Trigonometric Functions 3.4 The Chain Rule	CLO1/ CLO3	**Quiz 3** Thursday-18 (From 12:00 Pm to 12:30Pm)	4
5	Jul 21-25	3.4 The Chain Rule 3.5 Implicit Differentiation 3.6 Derivatives of Logarithmic Functions 3.9 Related Rates	CLO1/ CLO3		4
6	Jul 28- Aug 1	3.9 Related Rates 3.11 Hyperbolic Functions 4.1 Maximum and Minimum Values 4.2 The Mean Value Theorem	CLO2	**Major II** {2-8 and Ch 3} Thursday-1 (From 12:00 Pm to 1:00 Pm)	4
7	Aug 4-8	4.3 How Derivatives affect the Shape of a Graph 4.4 Indeterminate Forms and L'Hopital's Rule 4.4 Indeterminate Forms and L'Hopital's Rule 4.5 Summary of Curve Sketching	CLO2	**Quiz 4** Thursday-8 (From 12:00 Pm to 12:30Pm)	4
8	Aug 11-13	4.5 Summary of Curve Sketching 4.7 Optimization	CLO2		4
8	Aug-15 (Thursday)	**Final Exam**			

IV. Student Assessment & Teaching Strategies:

Domain	Assessment Task	Teaching Strategy
Knowledge		
Skills CLO1 : Calculate the limit of a function using appropriate techniques including l'hospital's Rule CLO2: Compute the derivative of functions by applying the rules of differentiation CLO3: Determine whether a function is continuous and/or differentiable CLO4: Find extreme values of functions CLO6 : Solve various real word problems using Calculus concepts	**Direct Assessment Tool** Quiz ,Exams Assignments **Indirect Assessment Tool:** Course Exit Survey	Lectures; Problem Solving sessions; Hands-on Practice; Examples presented in class; Mapping all the limit and differentiation techniques.
Competencies CLO5: Apply the concept of differentiation to sketch the graph of a function in a systematic way	**Direct Assessment Tool** Quiz, Exams Assignments **Indirect Assessment Tool** Course Exit Survey	Lectures Tutorials Problem Solving sessions Hands-on Practice

V. Course Requirements Two major exams; 4 quizzes; 8 online assignments; project or flipped classroom activity; Attendance & class participation.

VI. Schedule of Assessment

#	Assessment task*	Week Due	% of Total Assessment Score

1	First Major Exam	Week 3: Thursday, **11 Jul** (12-1 pm)	%20
2	Second Major Exam	Week 6: Thursday, **1 Aug** (12-1 pm)	%20
3	4 Quizzes	Distributed along the term	%10
4	Weekly Assignments	Throughout the semester	%5
5	Participation	Throughout the semester	%5
6	Final Exam		%40
7	Total		%100

VII. Learning Resources

A. References

Required Textbooks	Calculus: Early Transcendenatl Functions. 9th Edition by J. Stewart, D Clegg, S. Watson 1
Essential References Materials	1. Calculus for Business, Economics, life Sciences, and the Social Sciences By: Raymond Barnett, Michael Ziegler and Karl Byleen; 11th Edition. 2. Brief Calculus for the Business, Social & life Sciences By: Bill Armstrong and Don Davis; 3rd Edition.
Electronic Materials	Access to university LMS. We use the LMS to communicate with the students, Announcements and Grading are all conveyed to the students through this system. WebAssign : www.webassign.net . We use the WEBASSIGN to create online assignment and have a library of videos published by the author
Other Learning Materials	Slides and/or Videos are recorded and published via LMS or links are shared on moodle

B. Facilities Required- Classroom with projector and white board and smartboard and sometimes use of a lab.

C. EDU-HUB Platform

The platform will be used for all online education experiences at PSU. It supports a variety of methods to support the teaching and learning strategies across all disciplines. It also includes the LMS.

a. **Learning Management System** – It is required to make use of MOODLE. I will upload the syllabus and other course materials for your use as well as solutions to quizzes and Major exams.

VIII. Classroom Policies

A. Academic Integrity Policy (e.g., plagiarism or dishonesty)

"Plagiarism can be defined as unintentionally or deliberately using another person's writing or ideas as though they are one's own. Plagiarism includes, but is not limited to, copying another individual's work and taking credit for it, paraphrasing information from a source without proper documentation, and mixing one's own words with those of another author without attribution. In addition, buying a paper or project, or downloading a paper from the Internet, and submitting them as your own is also plagiarism. The penalty for academic dishonesty will bring course expulsion and failure, or even suspension" (Academic Integrity and Syllabus Acknowledgement Form).

All students are expected to submit their 'own' work and not the work of others.

B. Attendance Policy

Please, adhere to the following guidelines:

1. The University attendance policy will be strictly followed. In this course, the absence of **(16)** hours result in a Denied Notice (**DN**). (Refer to the student regulation handbook for further details.)

   Please Note: IT IS THE SOLE RESPONSIBILITY OF THE STUDENT TO KEEP SATISFYING their level of attendance. Otherwise, a "DN" grade will be granted automatically during any time in the semester without any notice from the e-register system.

2. **No make-up exams will be allowed except for university-accepted documents.** (Refer to the student regulation handbook for further details.)

C. Homework Submission Policy *(policy depends upon the course or the instructor)*

There is an online tool "WebAssign", which is a very helpful tool that should improve your performance in the course with all the supporting features available. Your instructor will send you all the details about how to register or use it. Online Homework will be given during the term using WebAssign. Your instructor will determine a deadline for submitting each homework. Five points will be assigned to the Homework, and two marks for the personal study plan practice that must be solved after each section. The due dates are announced in the assessment table. Your user name and password of the WebAssign account will be sent to you via email. It is the sole responsibility of the student to ensure that he/she has the Access Code and completed the homework before the stated deadlines. Please note that after the deadlines have passed, homework, quizzes, will not be opened for any reason.

IX. My Assumptions

Tips on how to enhance your problem-solving abilities:

- Attend all classes and participate in the class.
- Do all the homework assignments on time. It is impossible to understand math without solving problems.
- You are urged to practice (but not memorize) more problems than the ones assigned.
- You should always try to solve a problem on your own before reading the solution or asking for help.
- If you find it difficult to handle a certain type of problems, you should try more problems of that type.
- You are encouraged to solve some of the review problems at the end of each chapter.
- The practice you get doing homework and reviewing the class lectures will make exam problems easier and quicker to solve.
- Many videos are provided from the instructors of the course watch it and contact your instructor via email, if you have any concern.

Tips on how to be successful in Calculus I (Math 111)

It is expected that you should:

1. Attend all classes, pay attention, and do not hesitate to ask questions;
2. Spend about 4 hours each week working on homework suggested problems, reviewing the class lectures, reading the textbook, and studying for quizzes and exams;
3. Try to solve problems on your own before reading solutions or asking for help.
4. Consult your instructor immediately when you face a continuous difficulty in the course;
5. Complete all homework suggested problems every week, check the correctness of your work and understand the methods and principles they illustrated;
6. Master the designed course topics before each quiz and exam, and if necessary, complete additional problems beyond those assigned and consult other sources if you find the assigned problems and text are insufficient;
7. Write your solutions to homework, exams, and quizzes problems in an organized way;